Fermented Feeds Research

Lab Results – Nutrient Analysis Report

Farmer Research Network (FRN)

Results presented on a Dry Matter (DM) basis
Organizations: CRANE | MHAC | RFRN | RODI | SOFDI

Total Ash

Optimal Range

4%–12%

- In range
- Above range
- Below range

Total Ash Results by Organization

Total Ash Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Maurice Wasike — 10.4	Margret Muthoni — 5.2	Velindah Ouko — 3.8	Clara Laboso — 3.3	Robert Temba — 2.2
Philip Ouma — 11.1	Hellen Kandia — 5.8	Joyce Achieng' — 4.2	Linet Chepkoech Kering — 3.4	Roseline Olunga — 2.3
Dennis Onyango — 11.4	Eugine Chiveli — 6.0	Samson Ouma — 4.2	Vincent Wamalwa — 3.9	Pius Anyangu — 14.2
-	Eugine Chiveli — 6.1	Volen Otieno — 4.2	David Maiyo — 4.1	-
-	Eugine Chiveli — 6.7	Edwin Ouma Ochalla — 4.3	-	-
-	Amos Kwaram — 6.7	David Oluoch — 4.8	-	-
-	Jane Muluso — 7.0	Abaniibi B — 5.0	-	-
-	Eugine Chiveli — 7.2	Abaniibi A — 5.1	-	-
-	Esther Sombei — 7.3	Odongo Dennis — 5.6	-	-
-	Esther Viketi — 7.8	Witharaga — 5.9	-	-
-	Eugine Chiveli — 7.9	Milicent Odongo — 7.1	-	-
-	Henry Wafula — 8.6	Joyce Akinyi — 8.6	-	-
-	Wycliffe Ajungo — 8.9	-	-	-
-	Elizabeth Engevu — 10.0	-	-	-
-	Emmelda Nasimiyu — 10.3	-	-	-
-	Pamela Evans — 16.2	-	-	-

Legend: Gray = Below permissible range | Green = Within range | Red = Above permissible range

⚠️ Too Low

Low ash generally means low levels of essential minerals . There could be bone disorder and metabolic diseases

🔺 Too High

Excessive ash (above 30%) can cause kidney damage and bone/joint issues in growing birds. High ash reduces digestibility.
Can cause low available carbohydrate and proteins and also indicates contaminations(with sands and other impurities)

Calcium

Optimal Range

0.7%–4.5%

- In range
- Above range
- Below range

Calcium Results by Organization

Calcium Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Maurice Wasike — 1.0	Esther Viketi — 0.8	Velindah Ouko — 0.5	Clara Laboso — 0.5	Robert Temba — 0.5
Philip Ouma — 1.8	Jane Muluso — 0.8	Joyce Achieng' — 0.5	David Maiyo — 0.5	Roseline Olunga — 0.5
Dennis Onyango — 2.0	Elizabeth Engevu — 0.9	Samson Ouma — 0.5	Vincent Wamalwa — 0.7	Pius Anyangu — 4.0
-	Emmelda Nasimiyu — 0.9	Volen Otieno — 0.5	Linet Chepkoech Kering — 0.7	-
-	Hellen Kandia — 1.0	David Oluoch — 0.5	-	-
-	Eugine Chiveli — 1.0	Edwin Ouma Ochalla — 0.5	-	-
-	Eugine Chiveli — 1.0	Abaniibi B — 0.6	-	-
-	Esther Sombei — 1.1	Abaniibi A — 0.6	-	-
-	Eugine Chiveli — 1.1	Odongo Dennis — 0.7	-	-
-	Margret Muthoni — 1.1	Milicent Odongo — 0.8	-	-
-	Wycliffe Ajungo — 1.3	Joyce Akinyi — 0.9	-	-
-	Eugine Chiveli — 1.3	Witharaga — 0.9	-	-
-	Amos Kwaram — 1.4	-	-	-
-	Henry Wafula — 1.5	-	-	-
-	Eugine Chiveli — 1.5	-	-	-
-	Pamela Evans — 1.8	-	-	-

⚠️ Too Low

Low calcium leads to brittle/weak bones, thin eggshells, reduced laying, and deformed bones in young birds.

🔺 Too High

Excess calcium above needs can damage kidneys, especially in young chicks (0–16 weeks limit: 1%). Avoid dolomite lime.

Phosphorus

Optimal Range

0.22%–0.76%

- In range
- Above range
- Below range

Phosphorus Results by Organization

Phosphorus Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Maurice Wasike — 0.3	Eugine Chiveli — 0.3	Velindah Ouko — 0.2	Clara Laboso — 0.2	Robert Temba — 0.3
Dennis Onyango — 0.6	Hellen Kandia — 0.3	Abaniibi B — 0.3	Linet Chepkoech Kering — 0.3	Roseline Olunga — 0.4
Philip Ouma — 0.6	Amos Kwaram — 0.4	David Oluoch — 0.3	Vincent Wamalwa — 0.3	Pius Anyangu — 0.6
-	Eugine Chiveli — 0.4	Joyce Achieng' — 0.3	David Maiyo — 0.3	-
-	Margret Muthoni — 0.4	Abaniibi A — 0.3	-	-
-	Esther Viketi — 0.4	Joyce Akinyi — 0.3	-	-
-	Eugine Chiveli — 0.4	Witharaga — 0.4	-	-
-	Jane Muluso — 0.4	Samson Ouma — 0.4	-	-
-	Henry Wafula — 0.4	Odongo Dennis — 0.5	-	-
-	Emmelda Nasimiyu — 0.4	Milicent Odongo — 0.5	-	-
-	Eugine Chiveli — 0.5	Edwin Ouma Ochalla — 0.5	-	-
-	Esther Sombei — 0.5	Volen Otieno — 0.5	-	-
-	Eugine Chiveli — 0.6	-	-	-
-	Wycliffe Ajungo — 0.6	-	-	-
-	Elizabeth Engevu — 0.6	-	-	-
-	Pamela Evans — 0.7	-	-	-

Legend: Gray = Below permissible range | Green = Within range | Red = Above permissible range

⚠️ Too Low

Low phosphorus causes rubbery/pliable bones, bent ribs, swollen joints, thin-shelled eggs, and reduced growth.

🔺 Too High

Excessive phosphorus can interfere with calcium absorption and create mineral imbalances, stressing kidneys.

Crude Fat

Optimal Range

3%–8%

- In range
- Above range
- Below range

Crude Fat Results by Organization

Crude Fat Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Maurice Wasike — 2.8	Margret Muthoni — 3.3	Joyce Akinyi — 2.8	Linet Chepkoech Kering — 2.8	Robert Temba — 3.1
Dennis Onyango — 5.5	Hellen Kandia — 3.4	Samson Ouma — 3.7	Clara Laboso — 3.7	Pius Anyangu — 4.0
Philip Ouma — 6.1	Henry Wafula — 3.5	Velindah Ouko — 3.8	David Maiyo — 4.0	Roseline Olunga — 4.0
-	Esther Sombei — 4.0	Milicent Odongo — 4.0	Vincent Wamalwa — 4.5	-
-	Jane Muluso — 4.2	Edwin Ouma Ochalla — 4.0	-	-
-	Eugine Chiveli — 4.3	David Oluoch — 4.0	-	-
-	Emmelda Nasimiyu — 4.4	Witharaga — 4.4	-	-
-	Esther Viketi — 4.9	Odongo Dennis — 4.5	-	-
-	Eugine Chiveli — 5.1	Abaniibi B — 4.5	-	-
-	Eugine Chiveli — 5.3	Abaniibi A — 4.6	-	-
-	Amos Kwaram — 5.4	Volen Otieno — 5.1	-	-
-	Wycliffe Ajungo — 5.4	Joyce Achieng' — 5.1	-	-
-	Eugine Chiveli — 5.4	-	-	-
-	Pamela Evans — 5.9	-	-	-
-	Eugine Chiveli — 6.3	-	-	-
-	Elizabeth Engevu — 7.7	-	-	-

Legend: Gray = Below permissible range | Green = Within range | Red = Above permissible range

⚠️ Too Low

Low fat causes poor absorption of fat-soluble vitamins (A, D, E, K), forces the liver to work harder via lipogenesis, and reduces weight gain.

🔺 Too High

High fat (>8%) causes fatty liver syndrome, sudden death syndrome in hot weather, reduced egg production, and excess abdominal fat.

Crude Protein

Optimal Range

15%–18%

- In range
- Above range
- Below range

Crude Protein Results by Organization

Crude Prot Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Philip Ouma — 10.8	Hellen Kandia — 10.6	Milicent Odongo — 9.4	Vincent Wamalwa — 7.9	Robert Temba — 8.5
Maurice Wasike — 11.4	Margret Muthoni — 11.2	Joyce Akinyi — 11.6	David Maiyo — 8.5	Roseline Olunga — 11.7
Dennis Onyango — 16.6	Jane Muluso — 11.5	Velindah Ouko — 11.9	Clara Laboso — 9.1	Pius Anyangu — 12.8
-	Henry Wafula — 11.9	Samson Ouma — 12.2	Linet Chepkoech Kering — 9.5	-
-	Eugine Chiveli — 12.3	Edwin Ouma Ochalla — 12.5	-	-
-	Pamela Evans — 12.3	Witharaga — 12.7	-	-
-	Emmelda Nasimiyu — 12.4	David Oluoch — 12.8	-	-
-	Wycliffe Ajungo — 12.8	Odongo Dennis — 12.9	-	-
-	Esther Sombei — 13.0	Abaniibi B — 14.0	-	-
-	Eugine Chiveli — 13.6	Volen Otieno — 14.2	-	-
-	Eugine Chiveli — 13.7	Abaniibi A — 14.3	-	-
-	Amos Kwaram — 13.8	Joyce Achieng' — 14.5	-	-
-	Eugine Chiveli — 15.1	-	-	-
-	Elizabeth Engevu — 15.7	-	-	-
-	Esther Viketi — 15.8	-	-	-
-	Eugine Chiveli — 16.6	-	-	-

⚠️ Too Low

Low protein reduces weight gain, feed efficiency, egg production/size/shell thickness, and can weaken immunity.

🔺 Too High

High protein increases water intake 3% per 1% protein rise → wet litter → footpad issues. Strains kidneys; raises ammonia levels.

Crude Fibre

Optimal Range

5%–10%

- In range
- Above range
- Below range

Crude Fibre Results by Organization

Fibre Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Maurice Wasike — 8.9	Esther Viketi — 5.5	Milicent Odongo — 2.7	Clara Laboso — 2.7	Robert Temba — 1.8
Dennis Onyango — 13.1	Jane Muluso — 6.0	Velindah Ouko — 4.1	David Maiyo — 2.8	Roseline Olunga — 2.9
Philip Ouma — 14.8	Eugine Chiveli — 6.3	Volen Otieno — 4.2	Linet Chepkoech Kering — 5.0	Pius Anyangu — 6.0
-	Eugine Chiveli — 6.8	Samson Ouma — 4.3	Vincent Wamalwa — 5.1	-
-	Hellen Kandia — 7.3	Joyce Achieng' — 4.4	-	-
-	Eugine Chiveli — 7.5	David Oluoch — 4.5	-	-
-	Henry Wafula — 7.6	Edwin Ouma Ochalla — 4.6	-	-
-	Esther Sombei — 7.6	Odongo Dennis — 5.5	-	-
-	Amos Kwaram — 8.3	Witharaga — 6.2	-	-
-	Wycliffe Ajungo — 9.1	Abaniibi A — 6.7	-	-
-	Eugine Chiveli — 9.1	Abaniibi B — 6.8	-	-
-	Eugine Chiveli — 9.4	Joyce Akinyi — 9.4	-	-
-	Emmelda Nasimiyu — 12.0	-	-	-
-	Elizabeth Engevu — 12.9	-	-	-
-	Pamela Evans — 14.1	-	-	-
-	Margret Muthoni — 18.1	-	-	-

⚠ Too Low

Too little fibre leads to underdeveloped gizzard, higher feather-pecking and cannibalism risks, and increased pathogen colonization (Salmonella, Campylobacter).

🔺 Too High

Too much fibre dilutes feed energy (AME), increases digesta viscosity causing sticky droppings/wet litter. Excess coarse fibre can cause gizzard impaction. Reduce digestability of feeds

Digestible Energy

Optimal Range

2700 - 3200 kcal/kg

- In range
- Above range
- Below range

There is a strong negative correlation(-0.898**) between fibre and DE as expected. The readings are however extremely low Suggesting likely byproducts used, incorrect unit, or misinterpreted reading from lab.

Digestible Energy Results by Organization

Digestible Energy Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Philip Ouma — 354.6	Pamela Evans — 334.9	Joyce Akinyi — 370.3	Linet Chepkoech Kering — 412.1	Pius Anyangu — 367.6
Dennis Onyango — 358.0	Margret Muthoni — 350.7	Witharaga — 404.9	Vincent Wamalwa — 419.0	Robert Temba — 432.9
Maurice Wasike — 365.2	Emmelda Nasimiyu — 361.0	Abaniibi A — 407.7	David Maiyo — 425.8	Roseline Olunga — 433.1
-	Elizabeth Engevu — 376.8	Abaniibi B — 407.7	Clara Laboso — 428.1	-
-	Henry Wafula — 382.4	Odongo Dennis — 409.6	-	-
-	Eugine Chiveli — 383.0	Milicent Odongo — 412.2	-	-
-	Wycliffe Ajungo — 385.4	David Oluoch — 415.2	-	-
-	Esther Sombei — 391.3	Edwin Ouma Ochalla — 416.6	-	-
-	Eugine Chiveli — 393.1	Samson Ouma — 417.3	-	-
-	Hellen Kandia — 395.3	Velindah Ouko — 420.1	-	-
-	Amos Kwaram — 397.9	Joyce Achieng' — 424.1	-	-
-	Jane Muluso — 400.6	Volen Otieno — 424.5	-	-
-	Esther Viketi — 403.2	-	-	-
-	Eugine Chiveli — 406.2	-	-	-
-	Eugine Chiveli — 407.2	-	-	-
-	Eugine Chiveli — 409.0	-	-	-

⚠ Too Low

- leads to stunted growth, poor feed conversion ratios (FCR), and decreased body weight.
- Chickens cannot efficiently utilize dietary protein, leading to higher Nitrogen excretion.

🔺 Too High

- can reduce broiler weight gain and worsen the FCR

Total Carbohydrates

Optimal Range

60%–85%

- In range
- Above range
- Below range

Total Carbohydrates Results by Organization

Total Carb Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Dennis Onyango — 66.4	Pamela Evans — 65.6	Abaniibi A — 76.0	David Maiyo — 83.4	Pius Anyangu — 69.0
Philip Ouma — 72.0	Elizabeth Engevu — 66.6	Joyce Achieng' — 76.1	Vincent Wamalwa — 83.7	Roseline Olunga — 81.9
Maurice Wasike — 75.4	Eugine Chiveli — 70.5	Abaniibi B — 76.5	Clara Laboso — 83.9	Robert Temba — 86.2
-	Esther Viketi — 71.4	Volen Otieno — 76.5	Linet Chepkoech Kering — 84.3	-
-	Wycliffe Ajungo — 72.8	Witharaga — 77.0	-	-
-	Emmelda Nasimiyu — 72.9	Odongo Dennis — 77.0	-	-
-	Eugine Chiveli — 73.7	Joyce Akinyi — 77.0	-	-
-	Eugine Chiveli — 73.8	David Oluoch — 78.4	-	-
-	Amos Kwaram — 74.1	Edwin Ouma Ochalla — 79.2	-	-
-	Eugine Chiveli — 74.2	Milicent Odongo — 79.5	-	-
-	Esther Sombei — 75.7	Samson Ouma — 79.9	-	-
-	Henry Wafula — 76.0	Velindah Ouko — 80.5	-	-
-	Eugine Chiveli — 76.3	-	-	-
-	Jane Muluso — 77.3	-	-	-
-	Hellen Kandia — 80.2	-	-	-
-	Margret Muthoni — 80.3	-	-	-

Legend: Gray = Below permissible range | Green = Within range | Red = Above permissible range

⚠️ Too Low

Low carbohydrates can reduce energy availability and impair growth performance and feed conversion ratios.

🔺 Too High

Fermentation naturally transforms complex carbohydrates into simpler forms. High raw carbs in unfermented feed reduce digestibility.

NFE (Non-N Free Extract) Results by Organization

NFE (Non-N Free Extract)

Optimal Range

40%–80%

- In range
- Above range
- Below range

Non N Free Extract Dry by Organization

Sorted within organization

CRANE	MHAC	RFRN	RODI	SOFDI
Dennis Onyango — 53.4	Pamela Evans — 51.5	Joyce Akinyi — 67.6	Vincent Wamalwa — 78.7	Pius Anyangu — 63.0
Philip Ouma — 57.2	Elizabeth Engevu — 53.7	Abaniibi A — 69.3	Linet Chepkoech Kering — 79.2	Roseline Olunga — 79.1
Maurice Wasike — 66.4	Emmelda Nasimiyu — 60.1	Abaniibi B — 69.7	David Maiyo — 80.6	Robert Temba — 84.5
-	Margret Muthoni — 62.2	Witharaga — 70.7	Clara Laboso — 81.3	-
-	Eugine Chiveli — 63.0	Odongo Dennis — 71.5	-	-
-	Wycliffe Ajungo — 63.8	Joyce Achieng' — 71.6	-	-
-	Eugine Chiveli — 64.6	Volen Otieno — 72.1	-	-
-	Eugine Chiveli — 65.0	David Oluoch — 73.9	-	-
-	Amos Kwaram — 65.8	Edwin Ouma Ochalla — 74.6	-	-
-	Esther Viketi — 65.9	Samson Ouma — 75.6	-	-
-	Eugine Chiveli — 67.0	Velindah Ouko — 76.4	-	-
-	Esther Sombei — 68.2	Milicent Odongo — 76.9	-	-
-	Henry Wafula — 68.4	-	-	-
-	Eugine Chiveli — 70.1	-	-	-
-	Jane Muluso — 71.3	-	-	-
-	Hellen Kandia — 72.9	-	-	-

⚠️ Too Low

Very low NFE suggests highly fermented substrate. May indicate microbes have consumed most carbohydrates; check energy availability.

🔺 Too High

High NFE is common with cassava-based fermented feeds. Longer fermentation reduces NFE levels as microbes use sugars.

Summary & Key Takeaways

✅ Ash
Most farmers within range. Pamela Evans (MHAC) & Pius Anyangu (SOFDI) above — review feed sources.

⚗️ Phosphorus
All within acceptable ranges. Good mineral balance maintained across organizations.

💪 Crude Protein
Major concern — most farmers BELOW optimal range (16–24%). Increase protein-rich ingredients in fermented mix.

🌽 Carbohydrates
All farmers in acceptable range. Fermentation is successfully transforming carbohydrates.

🌽 Digestible Energy
All farmers have extremely low range. moderate us of byproducts and also increase the quantity of Maize and other alternative sources of Go feeds

🦴 Calcium
Several RFRN & SOFDI farmers below 1% — critical for layers and growers. Supplement calcium sources.

💧 Crude Fat
Most in range (3–8%). Maurice Wasike (CRANE), Joyce Akinyi & Linet C. Kering slightly low.-Can add some cooking oil or ripe avocado, peanut, oil seeds etc

🌿 Fibre
Mixed results. Philip Ouma & Dennis Onyango (CRANE) above range. Several RFRN & SOFDI below range. Reduce tough leaves and stems, put young agroforestry trees.

🔬 NFE
All within range. Fermentation is effectively converting complex carbs, supporting digestibility.

Recommendation: Focus on boosting Digestible Energy and crude protein levels across all organizations. Review fibre sources and calcium supplementation especially in RFRN and SOFDI.

Commercial Results by Organization

All

Optimal Range

—

In range

Above range

Below range

Commercial Feeds — Nutrient Analysis

All nutrients shown together · Color indicates bounds classification

Farmer / Feed	Fibre Dry	Crude Fat Dry	Crude Prot Dry	Non N Free Extract Dry	Total Carb Dry	Digestible Energy Dry	Fumonisins Dry
Kiminini growers mash	13.6	3.4	7.9	44.2	57.7	259.8	1525.3
RODI Ngeria Growers mash	22.0	3.0	8.3	51.1	73.1	287.4	528.4
Migori Pembe feeds layers mash	6.4	3.3	17.3	66.6	73.0	396.9	0.0
Migori Pembe feeds growers mash	7.4	5.0	14.4	57.6	65.0	361.9	1129.3
Chavakali ecofeeds growers mash	17.0	3.8	9.4	45.6	62.5	275.2	0.0

Legend: Gray = Below lower bound | Green = Within range | Red = Above upper bound

⚠️ Too Low

🔺 Too High